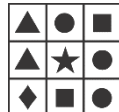


Canadian Math Kangaroo Contest

Part A: Each correct answer is worth 3 points

1. Together, mum Kangaroo and her son Jumper weigh 60 kilograms. Mum Kangaroo alone weighs 52 kilograms. How much does Jumper weigh?
 (A) 2 kilograms (B) 4 kilograms (C) 8 kilograms (D) 30 kilograms (E) 46 kilograms

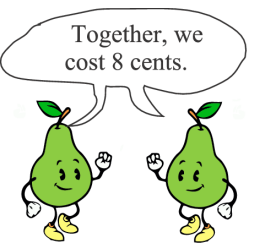
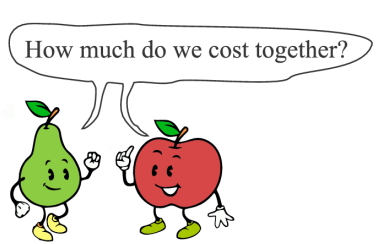
2. Karen cuts out one piece of this grid:



Which piece is the one she cut?

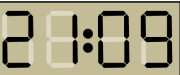
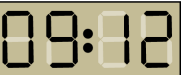
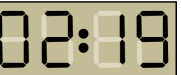
- (A)  (B)  (C)  (D)  (E) 

3. Yesterday was Sunday. What day is tomorrow?
 (A) Tuesday (B) Thursday (C) Wednesday (D) Monday (E) Saturday

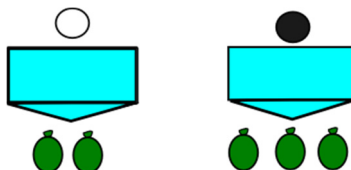
4.   
 (A) 5 cents (B) 6 cents (C) 7 cents (D) 8 cents (E) 9 cents

5. A digital clock shows the time 20:19. What will the clock show the next time it uses the same digits?



- (A)  (B)  (C)  (D)  (E) 

6. Amalia has a machine that gives her two balloons for one white token and three balloons for one black token as shown.



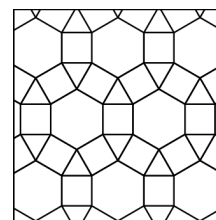
Amalia has one white and two black tokens:   .

How many balloons will Amalia get after inserting all her tokens into the machine?

- (A) 5 (B) 6 (C) 7 (D) 8 (E) 9

7. Which of the following geometric figures is not in this design?

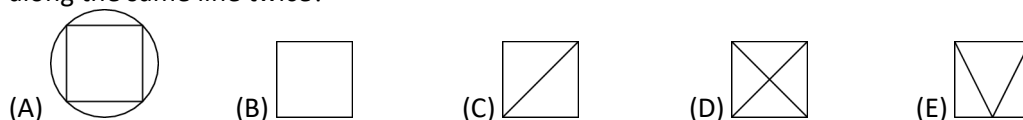
(A) triangle (B) square
(C) regular hexagon (D) regular octagon
(E) regular dodecagon



8. Which cloud contains four even numbers?

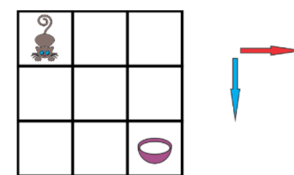
(A) (B) (C) (D) (E)

9. Which of the diagrams below cannot be drawn without lifting your pencil off the page and without drawing along the same line twice?



10. A cat and a bowl of milk are in the opposite corners of the board. The cat can only move as shown by the arrows. In how many ways can the cat reach the milk?

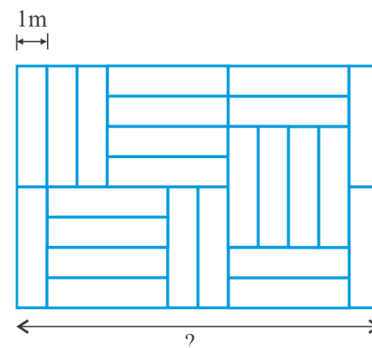
(A) 2 (B) 3 (C) 4
(D) 5 (E) 6



Part B: Each correct answer is worth 4 points

11. A floor is covered with identical rectangular tiles as shown. The shorter side of each tile is 1 m. What is the length of the side with the question mark?

(A) 6 m (B) 8 m (C) 10 m
(D) 11 m (E) 12 m



12. A train from Kang station to Aroo station leaves at 6:00 in the morning and passes by other three stations on the way, without stopping.



The numbers show the journey times between two stations, in hours. The train arrives at Aroo station at 11:00 at night on the same day. What is the journey time between Aroo station and the previous one?

(A) 2 hours (B) 3 hours (C) 4 hours (D) 5 hours (E) 6 hours

13. There are 10 camels in a Zoo. The camels are either bactrian (with two humps) or dromedary (with one hump). In total there are 14 humps. Find the number of bactrian camels in the Zoo.

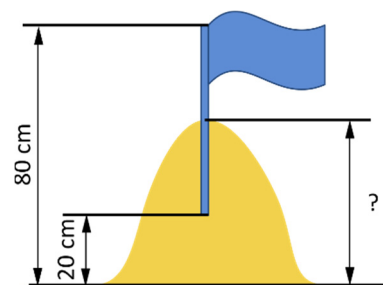
(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

14. Three squirrels Anni, Asia and Elli collected 7 nuts in total. Each squirrel collected a different number of nuts, but each collected at least one. Anni collected the least, Asia the most. How many nuts did Elli collect?

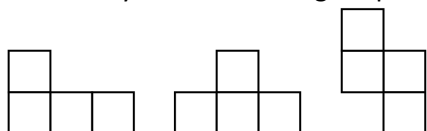
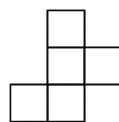
(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

15. Tim and Tom built a sandcastle and decorated it with a flag. They stuck half of the flagpole into the highest point of the castle. The upper tip of the flagpole was 80 cm above the ground, the lower tip was 20 cm above the ground. How tall was the sandcastle?

(A) 40 cm (B) 45 cm (C) 50 cm
(D) 55 cm (E) 60 cm



16. Dennis wants to remove one cell from the shape: How many of the following shapes can he get?



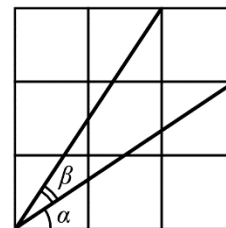
(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

17. Sara has 16 blue marbles. She can trade marbles in two ways: 3 blue marbles for 1 red marble or 2 red marbles for 5 green marbles. What is the maximum number of green marbles she can get?

(A) 5 (B) 10 (C) 13 (D) 15 (E) 20

18. Which of the following statements is certainly true for the marked angles in the given figure of nine small squares?

(A) $\alpha = \beta$ (B) $2\alpha + \beta = 90^\circ$ (C) $\alpha + \beta = 60^\circ$
(D) $2\beta + \alpha = 90^\circ$ (E) $\alpha + \beta = 45^\circ$



19. The triathlon consists of swimming, running, and biking. The biking is three-quarters of the total distance; the running is one-fifth; and the swimming is 2 km. What is the total distance of this triathlon, in km?

(A) 10 (B) 20 (C) 38 (D) 40 (E) 60

20. 60 apples and 60 pears are packed into boxes so that each box contains the same number of apples, and no two boxes contain the same number of pears. What is the largest possible number of boxes that can be packed in this way?

(A) 20 (B) 15 (C) 12 (D) 10 (E) 6

Part C: Each correct answer is worth 5 points

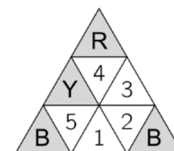
21. Each of the three shapes stands for a different number. The sum of the three numbers in each row is shown to the right of the row.

★	★	★	15
★	★	★	12
★	★	★	16

Which number does the ★ stand for?

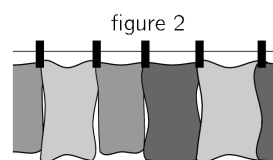
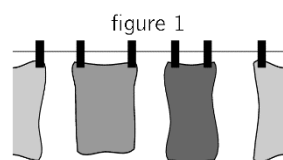
(A) 2 (B) 3 (C) 4 (D) 5 (E) 6

22. Mary has 9 small triangles: 3 of them are red (R), 3 are yellow (Y) and 3 are blue (B). She wants to form a big triangle by putting together these 9 small triangles so that any two triangles with an edge in common are different colours. Mary places some small triangles as shown in the picture. Which of the following statements is true after she has finished?



(A) 1 is yellow and 3 is red (B) 1 is blue and 2 is red
(C) 1 and 3 are red (D) 5 is red and 2 is yellow (E) 1 and 3 are yellow

23. Emil started to hang up towels using two pegs for each towel as shown in figure 1. He realised that he would have not enough pegs and began to hang up the towels as shown in figure 2.

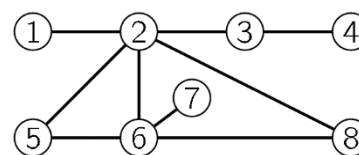


Overall, he hung up 35 towels and used 58 pegs.

How many towels did Emil hang up in the way shown in figure 1?

- (A) 12 (B) 13 (C) 21 (D) 22 (E) 23
24. Liam spent all his money buying 50 soda bottles at the store for \$1 each. He sells each bottle at the same higher price. After selling 40 bottles, he has \$10 more than he started with. He then sells all the remaining bottles. How much money does Liam now have?
- (A) \$70 (B) \$75 (C) \$80 (D) \$90 (E) \$100
25. Elisabeta had a large bag of 60 chocolates. She started by eating one 10th of them on Monday, then one 9th of the remainder on Tuesday, then one 8th of the rest on Wednesday, then one 7th on Thursday and so on until she eats half of the remaining chocolates from the previous day. How many chocolates does she have left?
- (A) 1 (B) 2 (C) 3 (D) 4 (E) 6

26. Prab painted each of the eight circles in the diagram either red, yellow or blue such that no two circles that are joined directly are painted the same colour. Which two circles are necessarily painted the same colour?



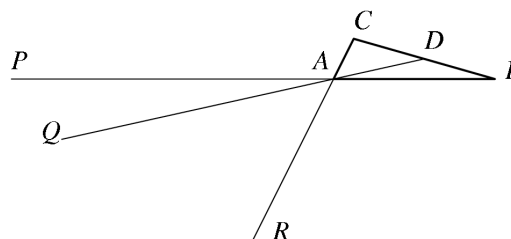
- (A) 5 and 8 (B) 1 and 6 (C) 2 and 7 (D) 4 and 5 (E) 3 and 6
27. Let a be the sum of all positive divisors of 1024 and b the product of all positive divisors of 1024. Which one of the following equalities is true?
- (A) $(a - 1)^5 = b$ (B) $(a + 1)^5 = b$ (C) $a^5 = b$
 (D) $a^5 - 1 = b$ (E) $a^5 + 1 = b$
28. How many planes pass through at least three vertices of a given cube?
- (A) 6 (B) 8 (C) 12 (D) 16 (E) 20
29. The numbers a , b , and c each have three digits, and for each number the first digit is the same as its last digit. Also $b = 2a + 1$ and $c = 2b + 1$. How many possibilities are there for the number a ?
- (A) 0 (B) 1 (C) 2 (D) 3 (E) more than 3

30. Given triangle ABC of area S , let D be the midpoint of BC . Take points P , Q , R on lines AB , AD , AC , respectively, as shown in the picture, and such that

$$AP = 2 \cdot AB, AQ = 3 \cdot AD \text{ and } AR = 4 \cdot AC.$$

What is the area of triangle PQR ?

- (A) S (B) $2S$ (C) $3S$
 (D) $\frac{1}{2}S$ (E) 0 (i.e. P , Q , R are collinear)



For training purposes only!

International Contest-Game Math Kangaroo Canada, 2019

Answer Key Parents

1	A B <u>C</u> D E	11	A B C D <u>E</u>	21	A B C D <u>E</u>
2	A B C D <u>E</u>	12	A B C <u>D</u> E	22	A B C D <u>E</u>
3	<u>A</u> B C D E	13	A B C <u>D</u> E	23	A B C <u>D</u> E
4	A B <u>C</u> D E	14	A <u>B</u> C D E	24	A <u>B</u> C D E
5	A B <u>C</u> D E	15	A B <u>C</u> D E	25	A B C D <u>E</u>
6	A B C <u>D</u> E	16	A B <u>C</u> D E	26	<u>A</u> B C D E
7	A B C <u>D</u> E	17	A <u>B</u> C D E	27	A <u>B</u> C D E
8	A B C D <u>E</u>	18	A <u>B</u> C D E	28	A B C D <u>E</u>
9	A B C <u>D</u> E	19	A B C <u>D</u> E	29	A B <u>C</u> D E
10	A B C D <u>E</u>	20	A B C <u>D</u> E	30	<u>A</u> B C D E